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PAPER

Scaling Laws for Template-Free Detection of Environmental Phase Modulation in Gravitational-Wave Signals

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Abstract

Environmental effects such as hierarchical triple motion introduce cumulative phase modulation in gravitational-wave signals through time-dependent line-of-sight acceleration. Whether such effects are observable depends on both deformation strength and signal-to-noise ratio (SNR). However, this relationship has not been quantified in a way that does not rely on a specific waveform model. Here we study how well such smooth time-warp distortions can be detected using template-free time–frequency representations. Instead of using reconstruction error alone, we examine trajectory-based statistics derived from the continuous wavelet transform, in particular the evolution of the frequency centroid. We find that environmental modulation can be detected using a single-sample statistic referenced to an isolated-binary distribution, without requiring matched templates. Across a grid of cumulative phase distortions $\Delta\phi$ and SNR, the detection performance collapses onto a single scaling parameter,

$$\Lambda = \Delta\phi \times \mathrm{SNR}.$$

The ROC-AUC follows a sigmoid curve in Λ with a clear transition region. For moderate distortions ($\Delta\phi \gtrsim 3\,\mathrm{rad}$), environmental modulation is detectable even at low SNR. For smaller distortions ($\Delta\phi \sim 1\,\mathrm{rad}$), detectability is noise-limited and emerges only at $\mathrm{SNR} \gtrsim 20$. This indicates that smooth environmental phase modulation is not generically absorbed by intrinsic waveform variability. Instead, detectability is set by a simple scaling between cumulative phase distortion and signal strength. This scaling relation provides a concrete reference point for template-free environmental searches in gravitational-wave data.

1 Introduction

The universe is filled with star systems made up of more than one star. Observation has shown that high mass stars, like O-type, are found predominately in multiple systems [1] and a good chunk of those exist in triples or higher, more so with increasing mass [2]. The systems with three stars are called hierarchical triple systems (HTS) and are important for gravitational wave detection.

Matched filtering using waveform templates representing binary systems have been used successfully for detection and parameter estimation [3]. However, HTS change gravitational waveforms systematically. For HTS motion, the inner binary about a tertiary companion induces a time-dependent line-of-sight acceleration (LOSA) [4, 5] and structured acceleration profiles can produce nontrivial phase evolution beyond constant-acceleration approximations [6, 7]. This can create phase modulation that accumulates over time producing cumulative phase drift. If in detection we assume the source is an isolated binary, but the system is an HTS, this phase drift may be absorbed into intrinsic parameters biasing estimations of mass, chirp rate, or spin. In other words, the LOSA signal stays inside the isolated binary envelope hidden through incorrect parameter estimation – the effect is degenerate.

Recent work has shown that template-free models learn structured time-frequency representations of compact binaries [8]. In template-based analyses, this degeneracy appears as biased parameter recovery. In a template-free setting, however, the question is different: does LOSA move the signal to a statistically distinct location in representation space?

Using time–frequency representations computed from continuous wavelet transforms (CWT) we show that LOSA induces coherent deviations in the trajectory of the power-weighted mean frequency of the signal. We introduce a single-sample, distribution-referenced statistic and evaluate its detectability across a controlled grid space of cumulative phase distortions $\Delta\phi$ and signal-to-noise ratio (SNR). We demonstrate that receiver operating characteristic (ROC) performance approximately collapses onto a one-parameter scaling relation governed by

$$\Lambda = \Delta\phi \times \mathrm{SNR}.$$

Performance remains near chance below a characteristic Λ, and rises rapidly above it. In particular, phase distortions $\Delta\phi \gtrsim 3\,\mathrm{rad}$ are detectable even at low SNR, but $\Delta\phi \sim 1\,\mathrm{rad}$ becomes observable only when $\mathrm{SNR} \gtrsim 20$.

This paper shows that LOSA distortions produce measurable shifts in centroid trajectories that follow predictable SNR-$\Delta\phi$ scaling. Their observability is governed by a scaling between cumulative phase shift and SNR. This scaling provides a physically interpretable baseline for template-free environmental searches in current and future GW detectors.

2 Waveform Model and Environmental Phase Modulation

2.1 Isolated Chirp Model

We consider a simplified inspiral-like chirp signal whose strain can be written as

$$h_{\mathrm{iso}}(t) = A(t)\cos[\phi_{\mathrm{iso}}(t)]\,, \tag{1}$$

where $A(t)$ is a smooth amplitude envelope and $\phi_{\mathrm{iso}}(t)$ is a monotonically increasing phase. The instantaneous frequency is

$$f_{\mathrm{iso}}(t) = \frac{1}{2\pi}\frac{d\phi_{\mathrm{iso}}}{dt}. \tag{2}$$

For concreteness, we employ a Gaussian-modulated chirp with frequency evolving from f_{start} to f_{end} over the signal duration. The precise functional form is not critical; what matters is that $f_{\mathrm{iso}}(t)$ is smooth and increasing, mimicking the qualitative structure of compact-binary inspiral.

2.2 Line-of-Sight Acceleration as a Time Warp

Environmental motion induces a detector-frame time reparameterization. If the source experiences line-of-sight acceleration, the observed signal becomes

$$h_{\mathrm{LOSA}}(t) = h_{\mathrm{iso}}(t + \Delta t(t))\,, \tag{3}$$

where $\Delta t(t)$ is a smooth, slowly varying time shift determined by the relative motion of the source.

In the constant-acceleration approximation,

$$\Delta t(t) = \frac{a_{\parallel}}{c}\frac{t^2}{2}, \tag{4}$$

where $a_{\parallel}$ is the line-of-sight acceleration and c is the speed of light. More general acceleration profiles yield correspondingly smooth $\Delta t(t)$ functions.

This deformation is purely kinematic. The intrinsic phase evolution remains unchanged, but it is evaluated at a shifted time coordinate. The strain-domain difference

$$\Delta h(t) = h_{\mathrm{LOSA}}(t) - h_{\mathrm{iso}}(t) \tag{5}$$

can be visually small even when the cumulative phase distortion is significant.

2.3 Cumulative Phase Distortion

The observable effect of LOSA is best characterized by the cumulative phase difference between the deformed and isolated signals,

$$\Delta\phi(t) = \phi_{\mathrm{iso}}(t + \Delta t(t)) - \phi_{\mathrm{iso}}(t). \tag{6}$$

For small time shifts,

$$\Delta\phi(t) \approx 2\pi f_{\mathrm{iso}}(t)\,\Delta t(t). \tag{7}$$

We define the total environmental phase distortion

$$\Delta\phi = \max_t |\Delta\phi(t)|\,, \tag{8}$$

which serves as a convenient one-parameter measure of deformation strength.

In our controlled experiments, $\Delta\phi$ is varied systematically while keeping the intrinsic waveform fixed. For example, $\Delta\phi = 3\,\mathrm{rad}$ corresponds to an end-of-signal time shift of order $\sim 10\,\mathrm{ms}$ for the fiducial chirp used here.

2.4 Effect on Instantaneous Frequency

Because the deformation is a time warp, the instantaneous frequency of the observed signal becomes

$$f_{\rm LOSA}(t) = \frac{1}{2\pi}\frac{d}{dt}\phi_{\rm iso}(t + \Delta t(t))\,. \tag{9}$$

Expanding to first order,

$$f_{\rm LOSA}(t) \approx f_{\rm iso}(t) + \dot{\Delta t}(t)\, f_{\rm iso}(t) + \Delta t(t)\, \dot{f}_{\rm iso}(t). \tag{10}$$

Thus LOSA induces a smooth deviation in the frequency trajectory, even when the strain-domain difference appears visually subtle. It is this trajectory-level distortion that we quantify in the subsequent sections.

To illustrate the structure of this deformation, we show in Fig. 1 a representative example with $\Delta\phi = 3\,$rad in the noise-free limit. Panel (a) overlays the isolated and LOSA-deformed strain signals. Although the cumulative phase shift is several radians, the strain-domain difference remains visually small. Panel (b) displays the corresponding instantaneous frequency trajectories extracted from the time–frequency centroid. The environmental modulation manifests as a smooth shift in the frequency trajectory, even when the strain difference is subtle.

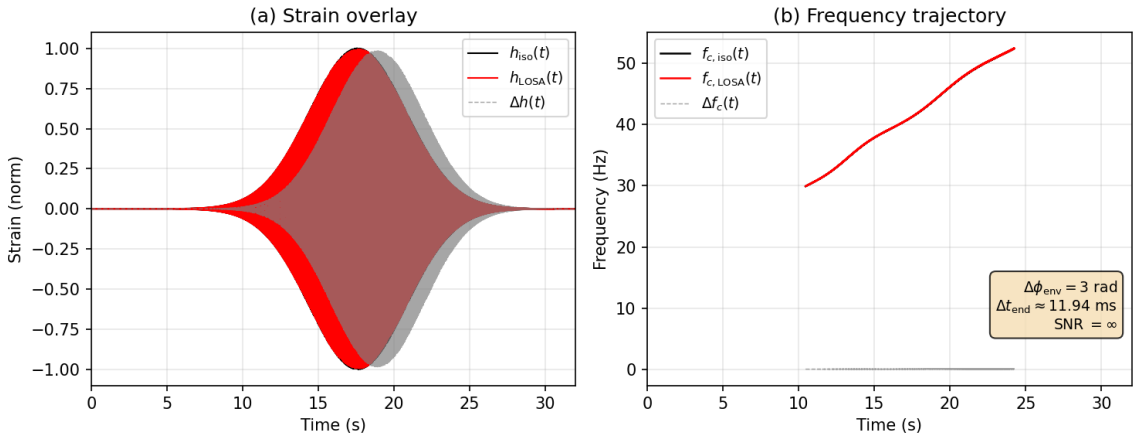

Figure 1. Illustration of environmental phase modulation for $\Delta\phi = 3\,$rad (noise-free). (a) Strain overlay showing the isolated chirp $h_{\rm iso}(t)$ and the LOSA-deformed signal $h_{\rm LOSA}(t)$. The strain-domain difference $\Delta h(t)$ remains small. (b) Instantaneous frequency trajectories obtained from the time–frequency centroid. The deformation induces a smooth shift in the frequency trajectory despite the small strain difference.

3 Experimental Design

The goal of this study is to determine when environmental phase modulation becomes statistically observable in noisy data. We treat detectability as a function of two independent controls: the cumulative environmental phase distortion $\Delta\phi_{\rm env}$ and the signal-to-noise ratio (SNR). Rather than relying on template matching or reconstruction error, we evaluate a single-sample statistic defined relative to a reference distribution of isolated signals.

3.1 Synthetic Signal Generation

We generate quasi-monochromatic inspiral-like chirp signals with frequency evolution spanning 12–65 Hz over a 32 s duration. The strain is tapered with a Gaussian envelope to suppress boundary artifacts. Environmental modulation is introduced as a smooth time reparameterization,

$$h_{\rm LOSA}(t) = h_{\rm iso}(t + \Delta t(t))\,, \tag{11}$$

where $\Delta t(t)$ corresponds to constant line-of-sight acceleration in the perturbative regime.

The deformation strength is characterized by the cumulative phase shift

$$\Delta\phi_{\rm env} = \max_t |\phi_{\rm LOSA}(t) - \phi_{\rm iso}(t)|\,. \tag{12}$$

Unless otherwise specified, signals are embedded in stationary Gaussian noise for statistical evaluation, while noise-free injections are used for visual diagnostics.

Detectability is evaluated on a controlled grid of

$$\Delta\phi_{\rm env} \in \{0.3,\, 1,\, 3\}\,\text{rad} \quad \text{and} \quad \text{SNR} \in \{5,\, 10,\, 20,\, 40\}.$$

Additional larger distortions are used for diagnostic checks. For each $(\Delta\phi_{\rm env}, \text{SNR})$ pair, isolated and LOSA-modulated samples are generated independently with distinct noise realizations. No pairing or oracle reference is used in the primary statistical evaluation.

3.2 Detection Statistic

Time–frequency representations are computed using the continuous wavelet transform. From the power spectrum $P(f,t)$, we extract the power-weighted frequency centroid

$$f_c(t) = \frac{\sum_k f_k P_{k,t}}{\sum_k P_{k,t}}, \tag{13}$$

restricted to gated time bins.

A reference trajectory $\mu_{\rm iso}(t)$ is constructed by averaging $f_c(t)$ over an ensemble of isolated training signals. For each evaluation sample (isolated or LOSA-modulated), we compute the scalar score

$$S_{fc} = \text{median}_t\, |f_c(t) - \mu_{\rm iso}(t)|\,. \tag{14}$$

This statistic measures deviation from the isolated centroid trajectory without access to the true isolated counterpart.

Detection performance is quantified using receiver operating characteristic (ROC) curves comparing isolated and LOSA samples. For each grid point, the area under the ROC curve (AUROC) is computed using S_{fc} as the ranking statistic.

3.3 Scaling Analysis

To assess whether detectability obeys a single-parameter control, we define the composite variable

$$\Lambda = \Delta\phi_{\rm env} \times \text{SNR}. \tag{15}$$

AUROC values from the full $(\Delta\phi_{\rm env}, \text{SNR})$ grid are analyzed as a function of Λ. This allows us to test for collapse onto a universal curve and to characterize the transition between degenerate and separable regimes via a sigmoid fit.

4 Results

4.1 Detectability Surface in SNR and Cumulative Phase Distortion

Figure 2 shows AUROC as a function of cumulative phase distortion $\Delta\phi_{\rm env}$ and signal-to-noise ratio (SNR). Each grid point represents an independent evaluation using the centroid-deviation statistic S_{fc}.

Several features are immediately apparent:

- For $\Delta\phi_{\rm env} \approx 0.3$ rad, detectability remains near chance across most SNR values.
- For $\Delta\phi_{\rm env} \approx 1$ rad, detection emerges at SNR $\gtrsim 20$.
- For $\Delta\phi_{\rm env} \approx 3$ rad, environmental modulation is detectable even at SNR ≈ 5.

These results indicate that detectability is jointly controlled by deformation strength and signal amplitude. Neither parameter alone determines observability.

4.2 Single-Parameter Scaling and Sigmoid Transition

To test whether the two-dimensional surface collapses onto a single control parameter, we define

$$\Lambda = \Delta\phi_{\rm env} \times \text{SNR}.$$

Figure 3 shows AUROC plotted against Λ for all measured grid points. The collapse is striking: data from different SNR and deformation combinations align along a common monotonic trajectory. This indicates that detectability is governed by the product $\Delta\phi_{\rm env} \times \text{SNR}$, rather than by either parameter independently.

To quantify the transition, we fit the scaling curve with a sigmoid model

$$\text{AUROC}(\Lambda) = \frac{1}{1 + \exp[-k(\Lambda - \Lambda_0)]}. \tag{16}$$

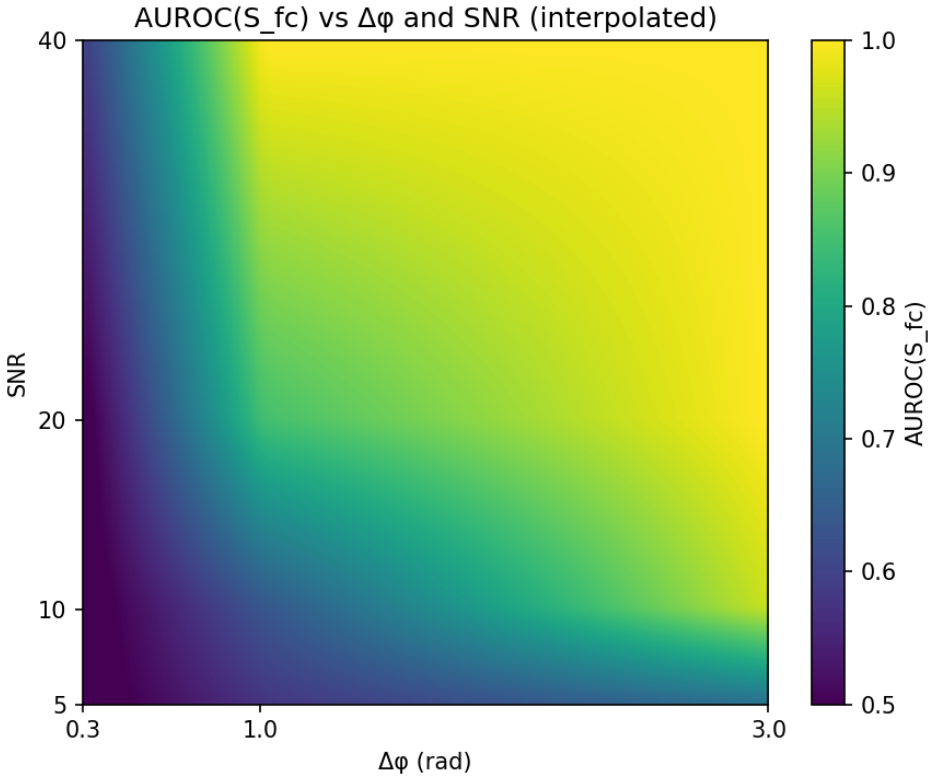

Figure 2. AUROC as a function of cumulative phase distortion $\Delta\phi_{\mathrm{env}}$ and SNR. Detectability increases with both deformation strength and signal amplitude.

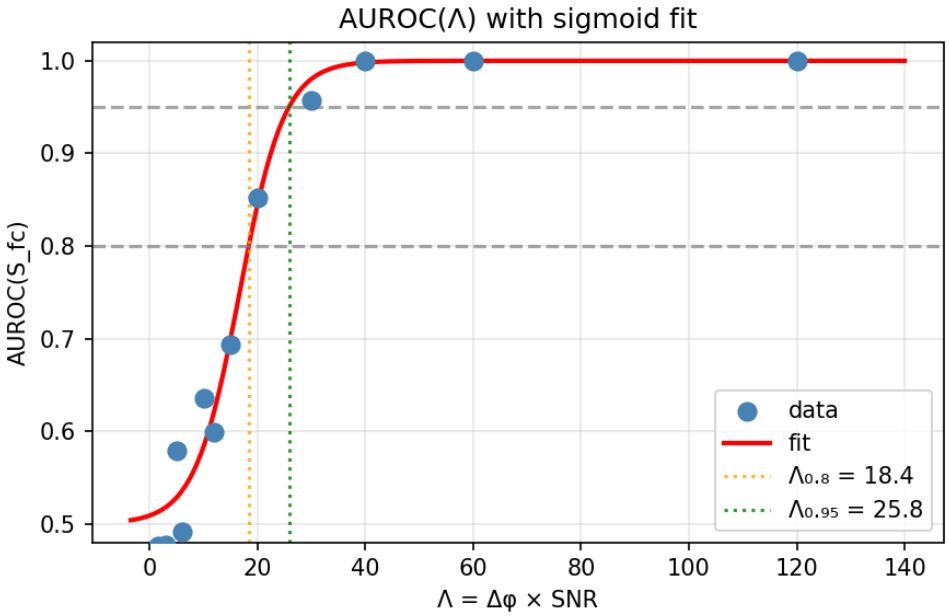

Figure 3. AUROC as a function of the composite parameter $\Lambda = \Delta\phi_{\mathrm{env}} \times \mathrm{SNR}$. Points from all $(\mathrm{SNR}, \Delta\phi)$ combinations collapse onto a single curve, demonstrating one-parameter scaling of environmental detectability. The red curve shows a sigmoid fit to the collapsed data. Vertical lines indicate characteristic threshold values corresponding to AUROC = 0.8 and 0.95.

Figure 3 shows the fitted curve along with threshold levels corresponding to AUROC = 0.8 and AUROC = 0.95.

The fitted parameters yield a well-defined transition scale Λ_0. Detection probability increases rapidly once Λ exceeds this threshold, indicating a sharp transition between noise-dominated and signal-dominated regimes.

4.3 Time-Local Behavior of the Centroid Shift

To verify that the centroid statistic reflects a physically meaningful deformation rather than a numerical artifact, Figure 4 shows the time evolution of the centroid difference

$$\Delta f_c(t) = f_{c,\mathrm{LOSA}}(t) - f_{c,\mathrm{iso}}(t)$$

for representative deformation strengths.

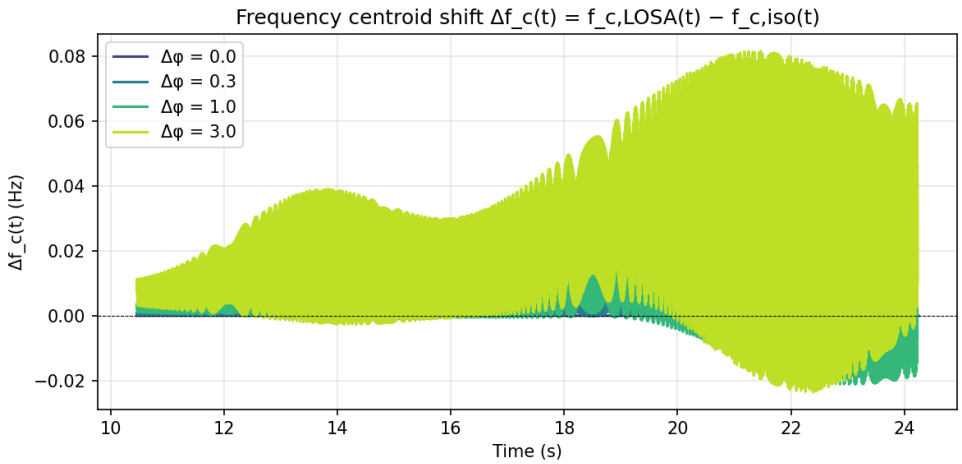

Figure 4. Time evolution of the centroid shift $\Delta f_c(t)$ for representative cumulative phase distortions. Larger $\Delta\phi_{\mathrm{env}}$ produces systematically larger centroid deviations.

The centroid shift grows smoothly with $\Delta\phi_{\mathrm{env}}$, confirming that the statistic tracks the expected time-warp deformation of the chirp trajectory.

4.4 Physical Interpretation

The scaling behavior can be understood intuitively. The cumulative phase distortion $\Delta\phi_{\mathrm{env}}$ sets the magnitude of the trajectory deformation, while SNR sets the inverse noise floor against which that deformation is measured. Their product therefore determines the effective signal-to-distortion ratio in the centroid trajectory.

Environmental phase modulation is not intrinsically degenerate with intrinsic waveform variability. Instead, observability is governed by a quantitative scaling relation:

$$\Lambda = \Delta\phi_{\mathrm{env}} \times \mathrm{SNR}.$$

Below the transition scale, noise fluctuations dominate centroid deviations. Above it, deformation-induced structure becomes statistically resolvable.

5 Discussion

5.1 Physical Interpretation of the Scaling Law

The results of Sec. 4 demonstrate that detectability of environmental phase modulation is governed by a simple composite parameter,

$$\Lambda = \Delta\phi \times \mathrm{SNR}. \tag{17}$$

Across a broad grid of cumulative phase distortions and signal-to-noise ratios, receiver operating characteristic performance collapses onto a single sigmoid curve in Λ.

This behavior admits a straightforward interpretation. The environmental effect produces a smooth time reparameterization of the waveform, which induces a systematic deviation in the time–frequency trajectory. In the perturbative regime, the magnitude of this trajectory deviation scales linearly with $\Delta\phi$. Measurement uncertainty in the trajectory, however, scales inversely with

SNR. The effective detectability is therefore controlled by the ratio of deterministic deformation amplitude to noise-induced variance, leading naturally to the product scaling $\Lambda = \Delta\phi\,\mathrm{SNR}$.

The observed sigmoid behavior reflects a transition between noise-dominated and deformation-dominated regimes. For $\Lambda \ll \Lambda_0$, environmental modulation is statistically indistinguishable from stochastic variation. For $\Lambda \gg \Lambda_0$, the deformation exceeds trajectory noise and separation approaches unity.

5.2 Noise-Limited Versus Physics-Limited Regimes

The detectability surface (Fig. 2) shows that environmental phase modulation is not fundamentally degenerate with intrinsic waveform structure. Rather, detectability is noise-limited at small $\Delta\phi$.

For moderate distortions ($\Delta\phi \gtrsim 3\,\mathrm{rad}$), detectability remains robust even at modest SNR. For smaller distortions ($\Delta\phi \sim 1\,\mathrm{rad}$), detectability improves sharply with increasing SNR, reaching near-perfect separation at $\mathrm{SNR} \gtrsim 40$.

Thus, the limiting factor in the weak-deformation regime is not representational degeneracy but statistical precision.

5.3 Implications for Template-Free Searches

The centroid-based statistic introduced here is single-sample and distribution-referenced; it does not rely on paired signals or matched templates. This demonstrates that template-free approaches can recover environmental signatures provided that the representation captures trajectory structure and that the composite parameter Λ exceeds threshold.

The scaling relation provides a quantitative baseline for environmental searches. Given an observed SNR, one can estimate the minimum cumulative phase distortion required for reliable detection. Conversely, for a hypothesized environmental deformation strength, the required SNR can be inferred.

5.4 Limitations and Future Work

The present analysis employs controlled, noise-injected chirp families and a simplified constant-acceleration model. More realistic triple-system dynamics will introduce time-dependent accelerations beyond quadratic time shifts. Future work should extend the scaling analysis to full post-Newtonian inspiral waveforms and realistic astrophysical distributions.

In addition, while the centroid statistic captures trajectory shifts cleanly, alternative representations sensitive to higher-order curvature changes may further lower detection thresholds.

Nonetheless, the central result remains: environmental phase modulation obeys a simple observability scaling law, and detectability is governed by the composite parameter $\Lambda = \Delta\phi \times \mathrm{SNR}$.

6 Implications for Current and Future Gravitational-Wave Detectors

The scaling relation identified in Sec. 4,

$$\Lambda = \Delta\phi \times \mathrm{SNR},$$

provides a simple criterion for environmental observability. From the sigmoid fit to the detectability surface, we find characteristic thresholds

$$\Lambda_{0.8} \approx 19, \qquad \Lambda_{0.95} \approx 27,$$

corresponding to AUROC values of 0.8 and 0.95, respectively. Environmental modulation becomes reliably detectable when Λ exceeds these values.

For ground-based detectors such as Advanced LIGO and Virgo, typical binary black hole events have network SNR $\sim$ 8–20. At $\mathrm{SNR} \approx 10$, the condition $\Lambda \gtrsim 20$ implies that cumulative phase distortions of order $\Delta\phi \gtrsim 2$ rad are required for robust detectability. For louder events with $\mathrm{SNR} \approx 20$, the threshold relaxes to $\Delta\phi \gtrsim 1$ rad. Thus, sub-radian environmental phase distortions are unlikely to be detectable in typical near-threshold ground-based events, while multi-radian distortions are in principle observable even at modest SNR.

The situation differs for space-based detectors such as LISA. Long-duration inspirals in the milli-Hz band can accumulate very high SNR over months to years of observation. In that regime, even relatively small cumulative phase distortions may satisfy $\Lambda \gtrsim 20$. The scaling law therefore suggests that environmental time-warp effects that are marginal in ground-based detectors could become observable in LISA-like observations, where both SNR and signal duration are substantially larger.

Importantly, these conclusions are derived without reliance on matched filtering or explicit waveform templates. The detectability thresholds arise purely from the statistical behavior of time–frequency trajectory deviations. While detailed astrophysical modeling is required to predict realistic values of $\Delta\phi$ for specific hierarchical configurations, the present framework provides a quantitative baseline for assessing whether a given environmental scenario is likely to be observable in a template-free setting.

In summary, environmental phase modulation is neither fundamentally invisible nor generically detectable. Its observability is governed by a simple scaling relation between cumulative phase distortion and signal strength. This scaling law offers a physically interpretable guide for future environmental searches in both current and next-generation gravitational-wave detectors.

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